Supplementary Material for: An Open-Source Pipeline for Spatiotemporal Traffic Congestion Analysis: Integrating HERE Traffic, OSMnx, and PySAL across Three Indonesian Cities

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S1. Spatial Weight Sensitivity Analysis

To confirm that the null global spatial autocorrelation finding reported in the main text is robust, we computed Moran's I under five spatial weight specifications for each city: K-nearest neighbours with $k \in \{4, 8, 12\}$ and distance-band weights at 500 m and 1000 m thresholds.

Table S1: Moran's I under alternative spatial weight specifications (overall mean jam factor). All 15 city–weight combinations yield non-significant results ($p > 0.05$), confirming the absence of global spatial autocorrelation.

City	KNN k=4	KNN k=8	KNN k=12	Dist 500 m	Dist 1000 m
Jakarta	0.001 (p=0.86)	0.003 (p=0.46)	0.005 (p=0.12)	−0.009 (p=0.14)	−0.005 (p=0.33)
Bandung	0.004 (p=0.74)	0.007 (p=0.35)	0.009 (p=0.20)	0.007 (p=0.63)	−0.000 (p=0.99)
Semarang	0.008 (p=0.67)	−0.004 (p=0.84)	0.003 (p=0.77)	0.003 (p=0.89)	−0.004 (p=0.88)

The distance-band weights at 500 m and 1000 m provide implicit correlogram information. Both distance thresholds yield non-significant Moran's I across all cities, indicating that spatial autocorrelation is absent at both local (neighbouring segments) and broader spatial scales.

S2. Period-Specific Moran's I

Global Moran's I was computed separately for each of the eight temporal periods to test whether spatial clustering emerges during specific times of day. Spatial weights use KNN with k=8.

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Table S2: Moran's I by temporal period (KNN k=8). No period–city combination reaches conventional significance ($p < 0.05$), confirming that the absence of global spatial clustering is not an artefact of temporal averaging.

Period	Jakarta	Bandung	Semarang
Night	0.000 (p=0.91)	0.002 (p=0.81)	0.004 (p=0.74)
Morning Peak	−0.001 (p=0.85)	0.009 (p=0.27)	−0.003 (p=0.86)
Morning Off-Peak	0.005 (p=0.19)	0.009 (p=0.28)	−0.008 (p=0.62)
Lunch Hours	0.001 (p=0.79)	0.003 (p=0.68)	−0.022 (p=0.14)
Afternoon Off-Peak	0.006 (p=0.10)	0.016 (p=0.058)	−0.001 (p=0.99)
Evening Peak	0.003 (p=0.46)	0.007 (p=0.35)	−0.004 (p=0.84)
Evening Off-Peak	0.004 (p=0.25)	−0.002 (p=0.81)	0.001 (p=0.89)
Late Night	−0.002 (p=0.61)	0.005 (p=0.51)	−0.018 (p=0.23)

The closest result to significance is Bandung's afternoon off-peak period (I=0.016, p=0.058), which does not survive conventional thresholds and would not survive Bonferroni correction for 24 simultaneous tests ($\alpha_{\text{adj}} = 0.05/24 = 0.002$).

S3. Getis-Ord Gi* Hot and Cold Spot Counts

Getis-Ord Gi* statistics were computed with KNN (k=8) binary weights and 999 permutations to triangulate the LISA findings reported in the main text.

Table S3: Getis-Ord Gi* hot and cold spot counts (evening peak, $p < 0.05$). Cold spots outnumber hot spots in all three cities, indicating that clusters of low congestion are slightly more prevalent than clusters of high congestion.

City	Hot Spots	Cold Spots	Not Significant	% Significant
Jakarta	660 (4.5%)	807 (5.5%)	13,082 (89.9%)	10.1%
Bandung	118 (3.8%)	178 (5.8%)	2,773 (90.4%)	9.6%
Semarang	39 (3.6%)	53 (4.9%)	984 (91.4%)	8.6%

Among hot spots, 95 (14.4%) in Jakarta, 15 (12.7%) in Bandung, and 2 (5.1%) in Semarang reach the stricter $p < 0.01$ threshold. The overall proportion of significant segments (∼9–10%) is consistent with the LISA results reported in the main text.

S4. Spatial Match Quality

The spatial join between HERE Traffic segments and OSMnx road network edges was validated by examining centroid-to-centroid match distances.

The high match quality (>85% within 50 m) confirms that the nearest-edge spatial join introduces minimal positional error, supporting the reliability of centrality–congestion correlation and capacity drop analyses that depend on this join.

Table S4: Spatial match quality: HERE Traffic segments to OSMnx edges. Median distances are under 20 m in all cities, and the 200 m filtering threshold retains 98–99% of matches.

City	n	Median (m)	% within 50 m	% within 100 m	% within 200 m
Jakarta	20,954	17.8	89.3%	97.7%	99.3%
Bandung	4,382	19.6	86.3%	96.4%	98.9%
Semarang	1,539	19.6	85.2%	94.6%	97.9%

S5. Spatial Regression Results

To properly account for spatial dependence, Spatial Lag and Spatial Error models (via PySAL's `spreg`) were estimated alongside OLS multiple regression. The dependent variable is mean evening-peak jam factor; independent variables are edge betweenness centrality and POI density within a 500 m buffer.

Table S5: Spatial regression results: jam factor $\sim$ betweenness centrality + POI density. Across all model specifications, R^2 values are negligible (<0.3%), confirming that spatial predictors explain virtually none of the congestion variance.

City	Model	R^2	β (betw.)	p	β (POI)	p	Spatial param.
Jakarta	OLS	0.003%	−0.048	0.635	0.000010	0.704	—
	Spatial Lag	0.006%	−0.337	0.168	−0.000005	0.899	$\rho = 2.731$
	Spatial Error	0.003%	−0.051	0.615	0.000009	0.708	$\lambda = 0.012$
Bandung	OLS	0.260%	0.270	0.005	−0.000010	0.803	—
	Spatial Lag	0.128%	0.231	0.064	−0.000000	0.996	$\rho = 0.350$
	Spatial Error	0.260%	0.269	0.005	−0.000010	0.817	$\lambda = 0.023$
Semarang	OLS	0.004%	−0.017	0.894	−0.000007	0.887	—
	Spatial Lag	0.013%	0.033	0.866	0.000055	0.706	$\rho = 2.568$
	Spatial Error	0.004%	−0.018	0.886	−0.000008	0.867	$\lambda = -0.034$

Bandung's OLS betweenness coefficient is statistically significant (p=0.005) but trivially small in magnitude ($R^2 = 0.26\%$). This significance vanishes in the Spatial Lag model (p=0.064), suggesting it is partially attributable to spatial dependence rather than a genuine centrality effect. The spatial autoregressive parameters (ρ, λ) are small, consistent with the weak global spatial autocorrelation reported in the main text and Table S1.

S6. Supplementary Figures

S6.1. Hotspot Maps

Figures S1–S3 show LISA cluster and Getis-Ord Gi* hotspot maps for the evening peak period across all three cities. The main text includes the Jakarta hotspot map (Figure 5); Bandung and Semarang maps are provided here.

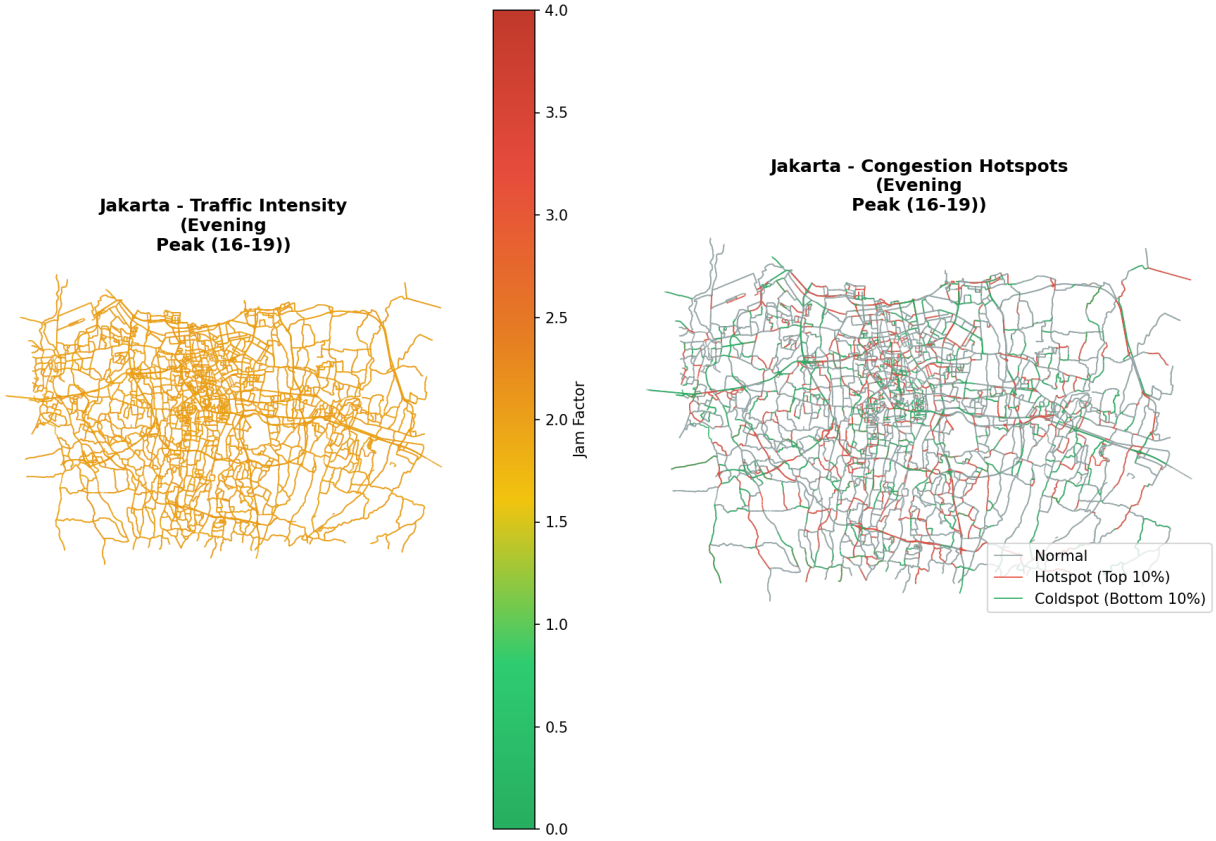

Figure S1: LISA cluster and Getis-Ord Gi* hotspot maps — Jakarta (evening peak). Hot spots concentrate around the inner toll-road corridors and central business district, while cold spots cluster in peripheral residential areas.

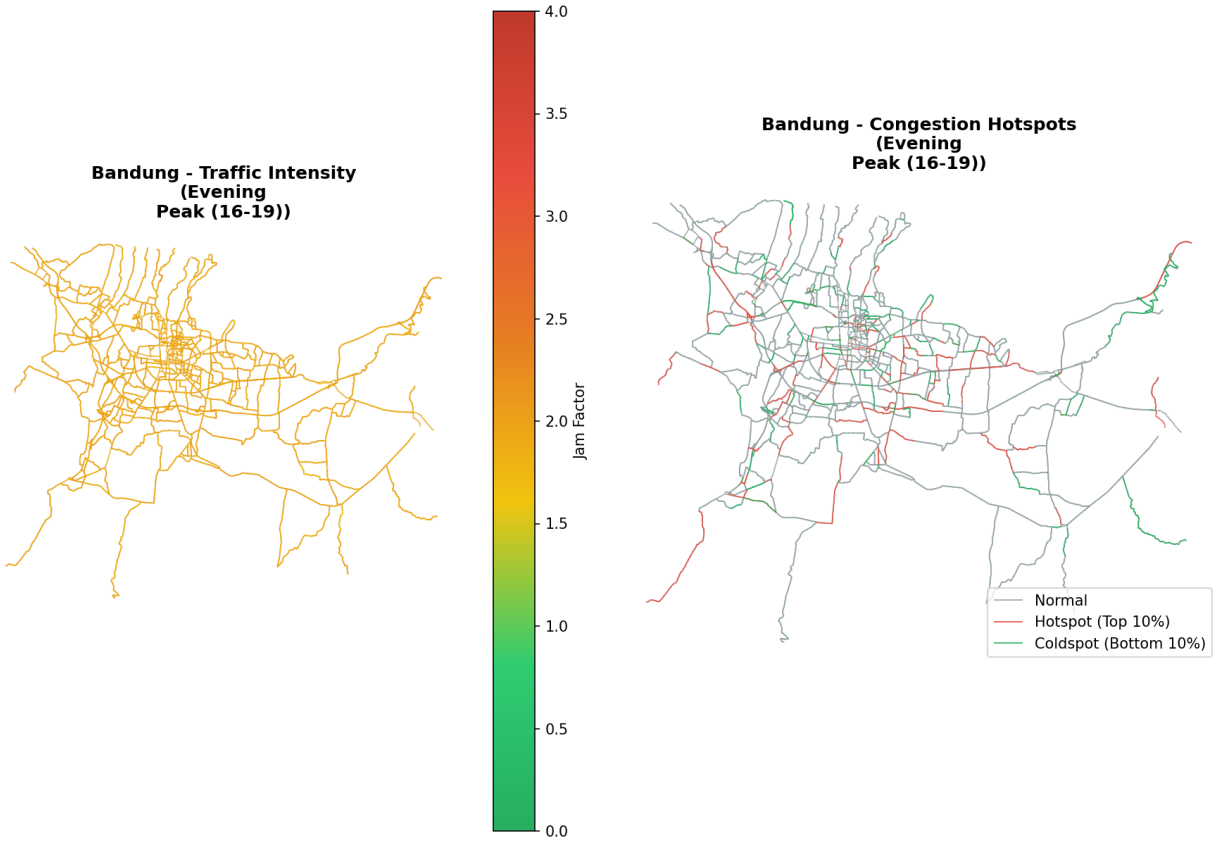

Figure S2: LISA cluster and Getis-Ord Gi* hotspot maps — Bandung (evening peak). Hot spots are concentrated along the northern industrial corridor (Jl. Soekarno-Hatta) and the Pasteur–Cipaganti commercial strip.

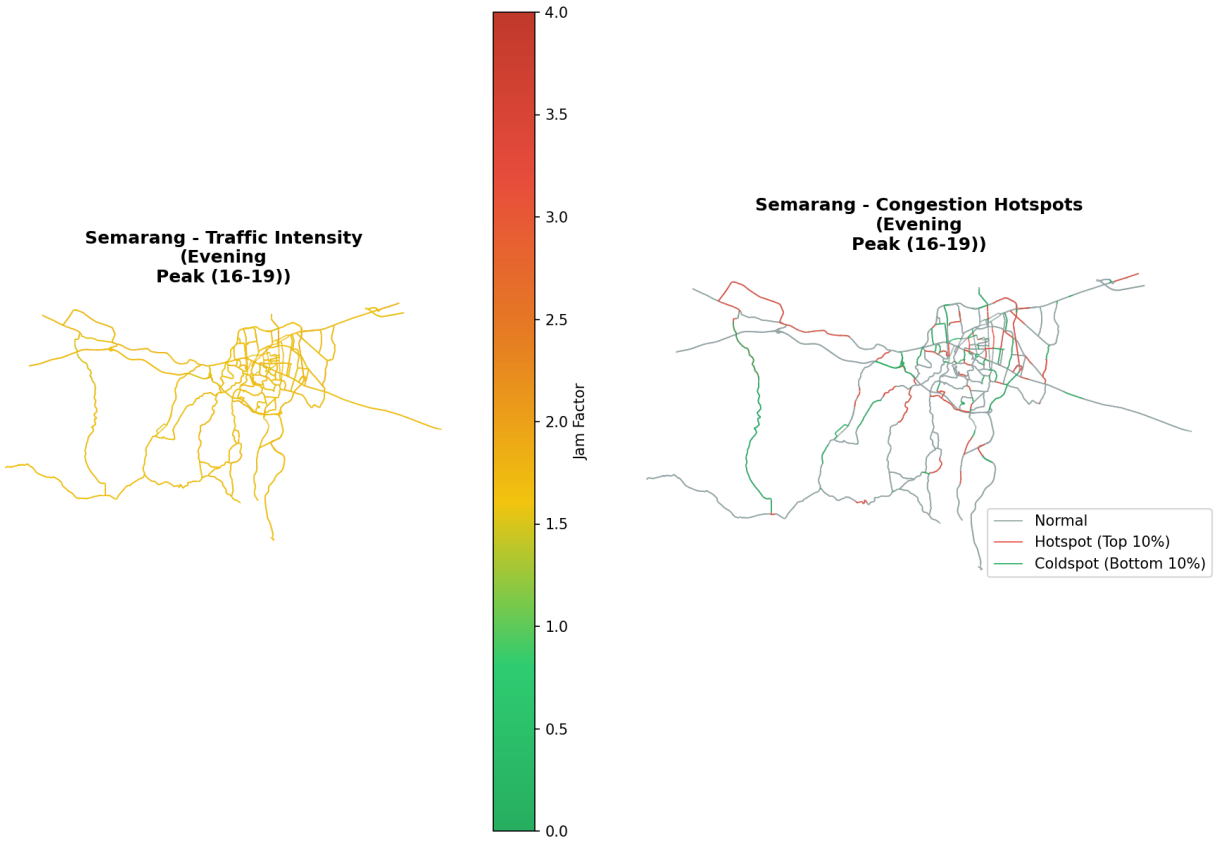

Figure S3: LISA cluster and Getis-Ord Gi* hotspot maps — Semarang (evening peak). Despite the smallest network (n=1,076), identifiable hot spots cluster around the Simpang Lima central roundabout and Jl. Pandanaran commercial corridor.

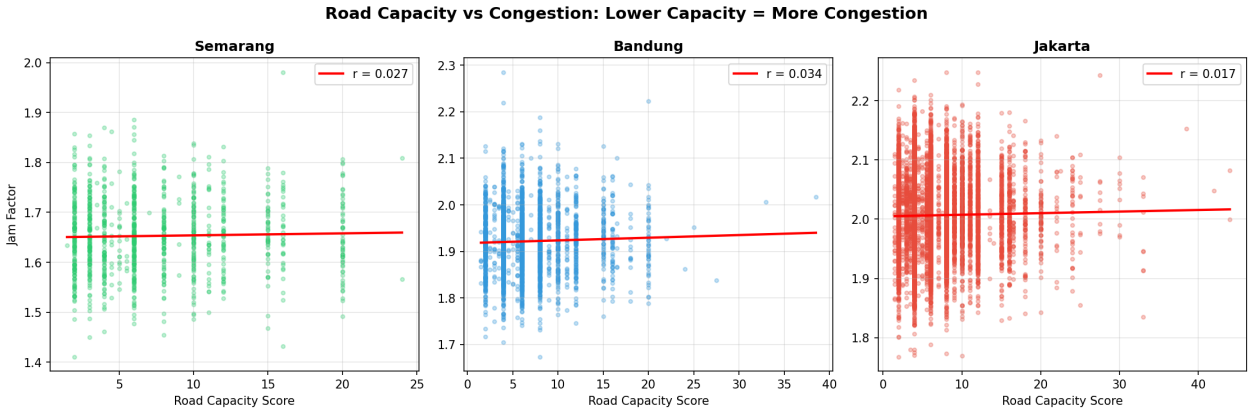

Figure S4: Scatter plots of road capacity (lanes × speed limit) versus mean evening-peak jam factor for all three cities. Points show individual road segments; LOESS curves confirm the flat capacity–congestion relationship.

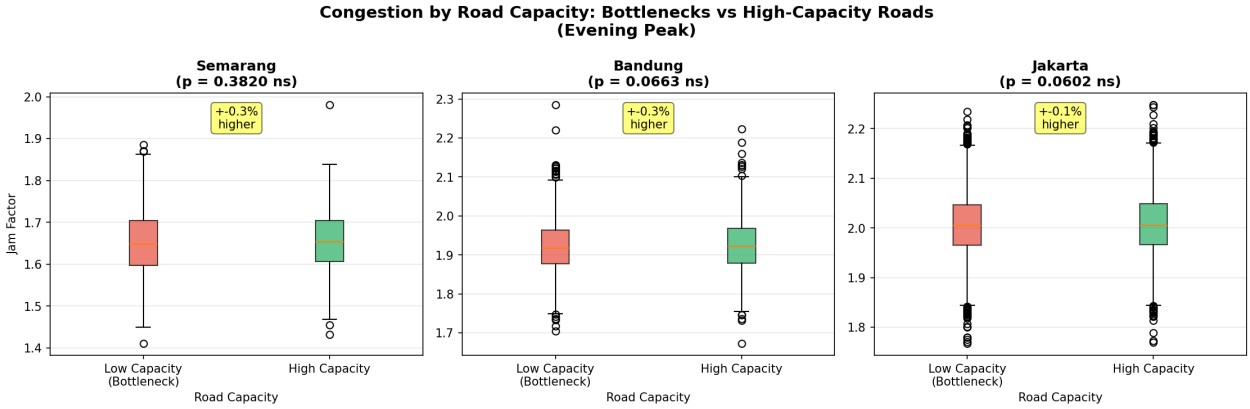

Figure S5: Bottleneck capacity comparison across cities. Box plots compare the distribution of road capacity attributes (number of lanes, speed limits) between bottleneck and non-bottleneck segments identified by the graph-based capacity drop detection algorithm.

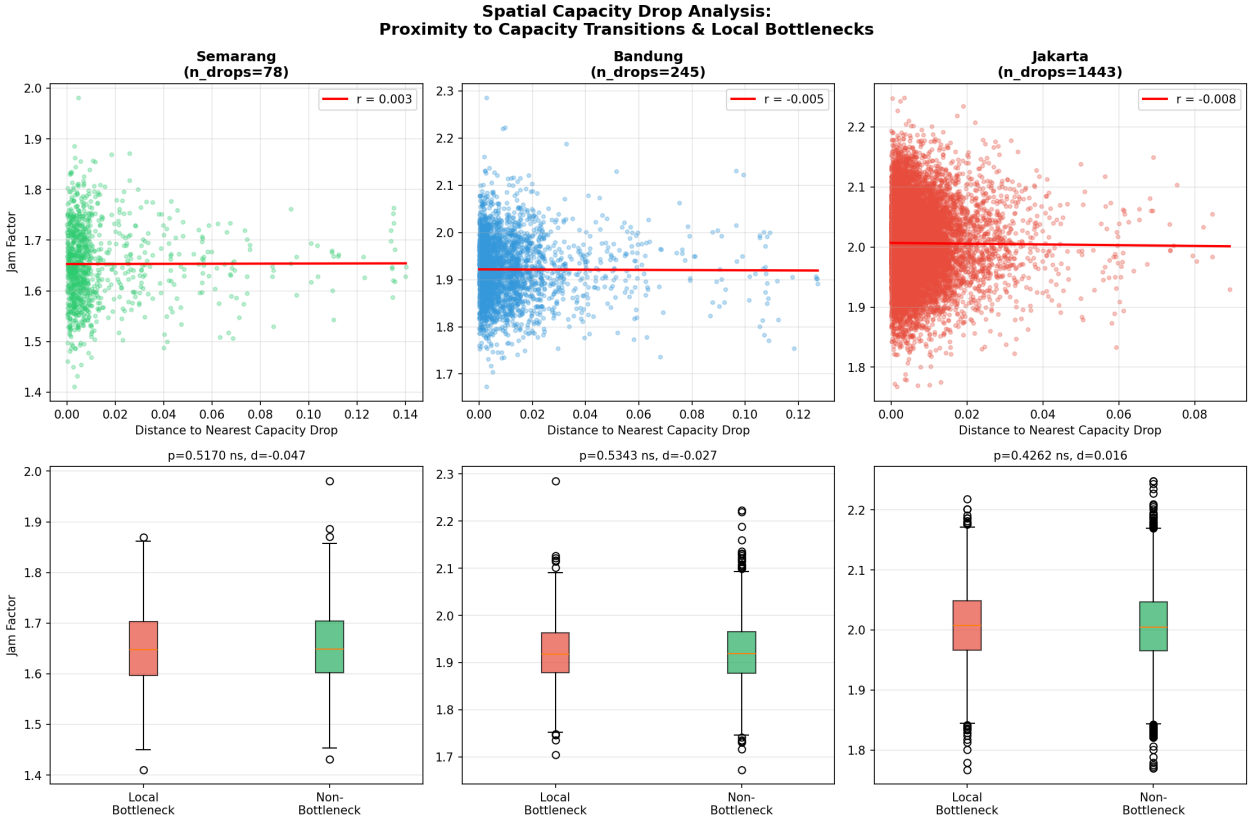

Figure S6: Spatial distribution of capacity drop segments across all three cities. Segments are classified by whether they exhibit a statistically significant capacity drop (jam factor increase when transitioning from high-capacity to lower-capacity downstream segments).

S6.2. Capacity–Congestion Scatter Plots

S6.3. Bottleneck Capacity Comparison

S6.4. Capacity Drop Spatial Distribution

S6.5. Street Orientation Analysis

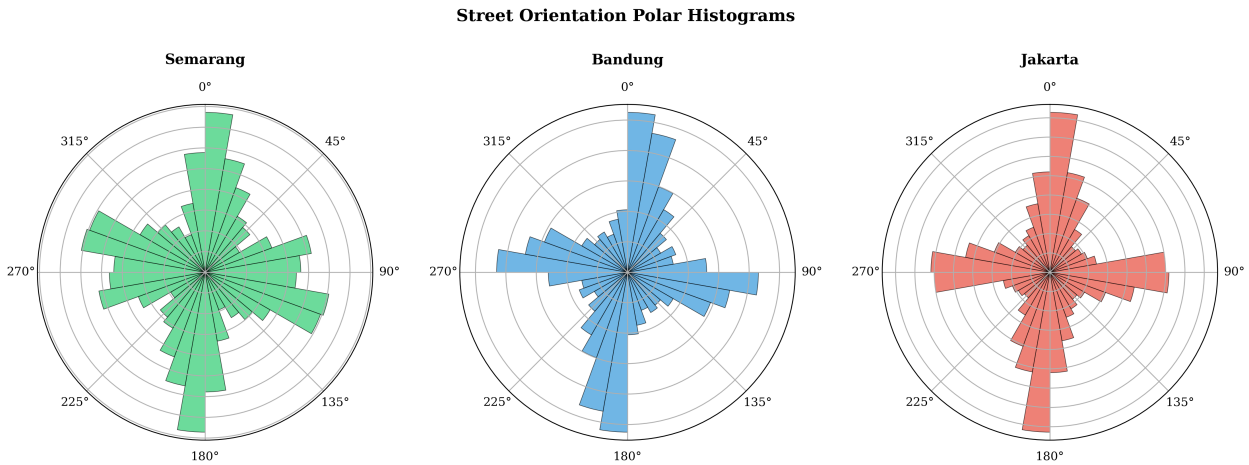

Figure S7: Street orientation polar plots for Jakarta, Bandung, and Semarang. Jakarta shows a pronounced grid pattern with dominant N–S and E–W orientations; Bandung and Semarang show more heterogeneous orientations reflecting their topographic constraints.